Color-Change Reaction

Teaching Lab - Student Procedure

Lab TL-04 | Grade level: middle school | Time: 10 minutes | Household materials only

Materials

Clear plastic cup, water, blue food coloring, baking soda, white vinegar, measuring spoon (1 teaspoon and 1 tablespoon), paper towels, safety goggles.

Safety

Wear safety goggles for the whole lab: vinegar stings if it splashes into your eyes. Do not taste any materials. Wipe up spills right away.

Procedure

Step 1: SAFETY - Put on safety goggles before handling any materials.
Step 2: Fill the clear cup halfway with water.
Step 3: Add 3 drops of blue food coloring to the water and stir with the spoon until the color is even.
Step 4: Add 1 teaspoon of baking soda to the cup with the measuring spoon and stir until the white powder dissolves.
Step 5: SAFETY - Keep goggles on. Slowly pour 2 tablespoons of vinegar into the cup and watch the mixture fizz and foam.
Step 6: When the fizzing stops, observe the color and foam, then pour the mixture into the sink and rinse the cup.

Questions

What gas makes the bubbles? Did the blue color change when the vinegar was added?